MedVisionQuery: An Agentic Multimodal System for Medical Visual Question Answering with Explanation

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Abstract

Medical visual question answering (VQA) over radiology images combines perception, question interpretation, answer synthesis and verification—steps that fail in different ways and are hidden inside a single model call. We present MedVisionQuery, an agentic system in which a Haystack Agent controller decides, case by case, how to look at the image, answer, verify the answer *against the image*, and whether to re-examine a specific region, revise, abstain or finish. Perception is question-aware, the answerer and the verifier both see the image, and a different, larger model verifies. We evaluate on 300 randomly sampled VQA-RAD test questions against a single-prompt baseline, a hand-designed workflow and a fixed-pipeline ablation, all on the same answering model, scoring with a strict rule-based scorer and a blinded LLM judge. The agentic loop is not more accurate than the single prompt (66.3% vs. 68.3% strict, 70.3% vs. 70.7% judged; McNemar $p = 0.52$ and $p = 1.00$) or the fixed pipeline ($p = 0.89$), and its larger verifier answering alone is significantly more accurate (72.0%, $p = 0.024$). Its revisions fix 23 wrong answers and break 5, but its abstentions withhold about as many correct ones. Its value is calibrated confidence: answers its check verifies are 79% correct, the rest 29%—at about 59$\times$ the cost of a single prompt. Code is available at `https://github.com/clarence-ck/med-vqa-rad-agentic`.

1 Problem Definition and Motivation

Medical VQA is a fundamentally multi-stage reasoning problem [1]. As established in the literature, a complete medical VQA system must address at minimum an image encoder, a question encoder, a multimodal fusion component, and an answer prediction module [1]—but these four components conceal distinct reasoning sub-problems that interact in clinically important ways. Given a radiology image and a clinical question, a system must separately solve: (1) *perception*—what is visible in the image that bears on the question; (2) *question interpretation*—what type of reasoning and answer is required, since different question types demand fundamentally different skills [2, 3]; (3) *synthesis*—how to integrate visual evidence into a grounded answer; (4) *verification*—whether the answer is actually supported by the visual evidence, a stage explicitly modelled in recent work as a "Confirm" module for cross-modal validation [5]; and (5) *control*—deciding, from what verification finds, whether to re-examine the image, revise the answer, abstain or stop.

These sub-tasks interact in non-trivial ways, which motivates making them explicit steps: what to look at depends on the question; the verifier must check cross-modal consistency between the image and the generated answer; and clinical safety requires explicit guardrails, abstention and disclaimers when confidence is low.

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Existing work: classification approaches. Early and still-prevalent medical VQA methods frame the task as multi-class classification over a fixed answer vocabulary [1]. The VQA-Med challenge [6] established the dominant paradigm: an image encoder and a question encoder produce fused features passed to a classification head. MEVF [7] extended this with meta-learning (MAML) and a denoising autoencoder (CDAE), feeding image features into a bilinear attention network (BAN) followed by a classifier. PMC-VQA [8] proposed MedVInT-TE, similarly mapping image–question pairs to answers via an encoder and classification head. Classification approaches are computationally efficient on close-ended questions but constrained to answers seen during training and perform poorly on open-ended questions requiring free-form responses.

Existing work: generation approaches. To handle open-ended questions, a parallel line of work adopts autoregressive generation. Bazi et al. [9] proposed a full transformer encoder–decoder architecture demonstrating gains on VQA-RAD and PathVQA. PMC-VQA's MedVInT-TD [8] further demonstrated that an LLM decoder can match or exceed classification-based performance when fine-tuned on large biomedical corpora. LLaVA-Med [10], accepted at NeurIPS 2023, adapted the general-domain LLaVA architecture to biomedicine via curriculum learning on PMC-15M figure–caption pairs and GPT-4-generated instruction-following data. RadFM [11] scaled this into a generalist radiology foundation model capable of VQA, report generation, and rationale diagnosis. R-LLaVA [12] extended LLaVA-Med by injecting clinician-annotated regions of interest into the CLIP encoder, improving spatial and contextual comprehension.

Existing work: hybrid approaches. Q2ATransformer [2] proposed a unified transformer framework introducing learnable candidate answer embeddings as a decoder, querying the fused image–question representation for each answer class. Although framed as classification, candidate answer embeddings interact through transformer attention, incorporating aspects of generative reasoning. This hybrid achieves state-of-the-art performance on VQA-RAD and PathVQA.

The gap that remains. Despite this progress, *all* prior approaches—whether classification [6–8], generation [9–12], or hybrid [2]—treat the task as a single end-to-end model invocation, making it impossible to localise failures, adapt the reasoning process to what the image shows, or enforce clinical safety constraints. Even recent works that acknowledge the need for verification [5] or knowledge grounding [4] address these as training objectives within a single model, rather than as explicit, auditable steps. MedVisionQuery fills this gap with an agent controller that coordinates question-aware perception, answering and image-grounded verification, deciding per case whether to re-examine the image, revise, abstain or finish, with structured failure handling.

We target the **VQA-RAD** benchmark [13], a clinically generated radiology VQA dataset; its Hugging Face release contains 2,244 image–question pairs (1,793 train / 451 test) across close-ended and open-ended question types. We evaluate on a seeded, stratified sample of 300 test questions; no training data is used.

2 Agent Architecture

Controller loop. A fresh Haystack `Agent` is created per request (Figure 1). Its five tools are bound to per-request working memory holding the image, the question analysis, and every look, answer and check; the controller sees only short text results and chooses the next call. Unlike the fixed-pipeline ablation, the tool sequence depends on what verification finds: on the 300 evaluation cases the controller finished directly in 186 cases, re-examined a region in 57, revised its answer in 35 and abstained in 22. Budgets cap the loop at two focused re-looks, three answers and 16 steps, with a safety net that checks the latest answer if the controller stalls.

Roles. The *controller* is the planner: it chooses every next step from the results so far. The per-request *working memory* (image, question analysis, looks, answers, checks) is the memory; the tools read from and write to it. `analyze_question` (task understanding) classifies the question type and expected answer format. `look(focus)` (perception) examines the image *with the question in mind*; with a focus, it zooms: the named region is located, cropped and enlarged, and examined beside the full image; the first look rejects non-radiology images. `answer(guidance)` (executor) answers from the image plus all observations, applying the verifier's guidance on revisions. `check` (evaluator, a larger model) has two parts. A *blind reading*, made once per episode: the verifier answers from the

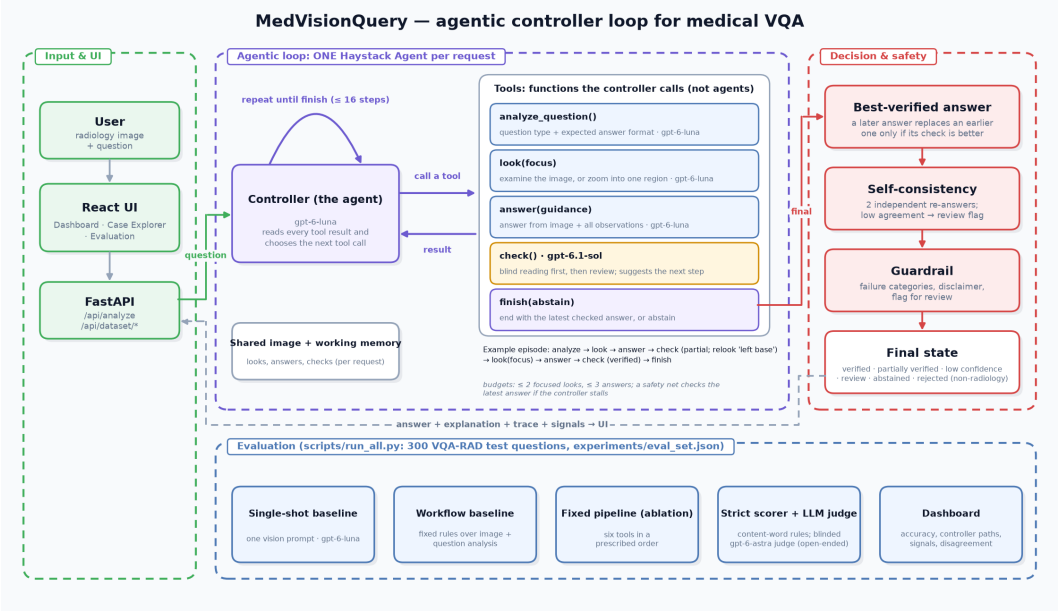


Figure 1: MedVisionQuery. One Haystack Agent (the controller) calls five tools in a loop, reading each result before choosing the next: question analysis, question-aware perception (`look`), answer synthesis, image-grounded verification (`check`) and `finish`. The tools are not agents; `check` only recommends a re-look, a revised answer, abstention or acceptance, and the controller decides. A decision-and-safety stage keeps the best-verified answer, measures self-consistency and applies guardrails.

image and question alone, without the proposed answer or the looker's notes. A *review*: it checks the answer against the image (grounding, contradictions, format) and returns a status (verified, partially verified, low confidence, unsupported), the issues found and a next step—*accept*, *relook* with a focus, *reanswer* with guidance, or *abstain*; the guidance may include a suggested corrected answer. An answer is verified only if the review passes and the blind reading agrees (by word rules, or for open-ended answers by meaning, judged by the verifier); the controller learns only whether they agree. **`finish(abstain)`** ends the episode; a normal finish is refused while the latest answer is unchecked.

Model and tool choices. All roles use OpenAI models, sized by need and configurable per role. The controller and question analysis make simple decisions (`gpt-6-luna`, no/low reasoning effort); perception and answering use `gpt-6-luna` with medium effort, and the answerer always sees the image, because a text description loses the visual detail many questions depend on. Verification uses a different, larger model (`gpt-6.1-sol`) so that the answering model never verifies itself; since its guidance can suggest answers, we also score it answering alone as a reference. Haystack's `Agent`/`Tool` abstraction provides the tool-calling loop and exit conditions; keeping the image in per-request memory rather than in the controller's context keeps the controller cheap.

Baselines and ablation. The *single-shot baseline* sends the image and question in one prompt. The *workflow baseline* is a fixed chain—a question-agnostic image description, question analysis, then hand-written rules per question type. The *fixed-pipeline ablation* is a controller instructed to call six tools in a prescribed order—`interpret_image`, `analyze_question`, `filter_image_context`, `route_difficulty`, `reason_answer`, `verify_answer`—re-running the sequence once when verification flags a fixable problem. It uses the same models but a question-agnostic image description, no zoom, no blind check and a text-only verifier, so comparing against it measures the agentic design as a whole, not control alone. A demo UI (Section 6) exposes all systems.

Table 1: Accuracy (%) on 300 random VQA-RAD test questions (same answering model for all systems). Open-ended accuracy is strict/judged. Time is per-case latency in seconds (mean/median), measured with four cases running concurrently; the whole run of all four systems took 71 minutes of wall-clock time. Cost is the mean API cost per case.

System	Strict	Judged	Close (169)	Open (131)	Time (s)	Cost
Single-shot baseline	68.3	70.7	81.7	51.1 / 56.5	4.7 / 3.8	\$0.0002
Workflow baseline	32.7	34.0	46.2	15.3 / 18.3	9.7 / 9.4	\$0.0005
Agentic loop	66.3	70.3	78.7	50.4 / 59.5	51.7 / 36.6	\$0.0135
Fixed pipeline (ablation)	67.0	68.7	77.5	53.4 / 57.3	36.5 / 29.6	\$0.0066

3 Multimodal Grounding

The system uses two modalities, the radiology image and the text question, and they interact at every step that determines the answer: the question (and any focus) steers what `look` examines; `answer` and `check` receive the image directly alongside the text observations, because a text description of the image loses the visual detail they depend on; only question analysis and the controller work from text. The evaluation's LLM judge compares answers with the reference as text. Because the verifier sees the image, its "explanation grounded in evidence" and "no contradictions" checks are genuine cross-modal agreement checks.

4 Control and Failure Handling

Detection. Image-grounded verification (five checks and a status), seven failure categories (*unsupported claim*, *low confidence*, *visual contradiction*, *partially verified*, *question–answer mismatch*, *generation unstable*, *parser failure*) and an eighth, *self-inconsistent*, from resampling. **Recovery.** Instead of re-running the whole pipeline, the controller zooms into the region the verifier names or revises the answer with the verifier's guidance. When the blind reading disagrees, the controller is told only *that* it disagrees, never the blind answer, so the loop must resolve it with its own model. A later answer replaces an earlier one only if its check is strictly better, because focused re-looks can over-read. **Self-critique.** After finishing, the answerer is re-asked independently twice; low agreement adds a failure category and flags the case for review. **Fallback.** The controller may abstain (reported as an abstention, never coerced into "no"); a guardrail adds a disclaimer and a review flag for low-confidence, unsupported, high-severity or self-inconsistent cases; non-radiology or unreadable images are rejected before any answer is produced.

5 Evaluation

Setup. 300 VQA-RAD test questions drawn at random, keeping the split's close/open-ended mix (seed 42; 169 close-ended, 131 open-ended); all four systems run on the same cases with the same answering model. *Strict scoring*: yes/no answers must start with the correct word; binary terms must name the correct term and not its opposite; counts must match; open-ended answers must contain every content word of the reference (aliases and plurals folded; $\geq$60% for references longer than three words), with left/right conflicts and denials counted wrong and embedding similarity [14] only as a strict fallback. Because word matching both rejects correct paraphrases and accepts vague answers that contain the reference word, a *blinded LLM judge* (`gpt-6-astra`, a model that neither answers nor verifies in any system; answers shuffled and anonymised) scores every open-ended answer as a second, independent measure; a 30-verdict spot check agreed with 29 credit decisions (the other, lenient, on a workflow answer). Paired comparisons use the exact McNemar test; 95% intervals are about $\pm$5.3 points. The baseline and agentic loop were re-run on 100 cases to measure stability. API cost of the evaluation: \$8.62 (\$6.25 runs, \$1.28 repeats, \$1.09 judge). Wall-clock time: 71 min for the main run (four cases at a time), 83 min for the sequential repeats and 10 min for the judge.

Results. Table 1. The agentic loop and the single prompt are statistically indistinguishable (27 vs. 33 discordant cases, $p = 0.52$; judged 30 vs. 31, $p = 1.00$), as are the loop and the fixed pipeline ($p = 0.89$; judged $p = 0.56$); all three far exceed the workflow. The verifier model

Table 2: Left: accuracy (strict, %) by the path the controller took. Right: agentic accuracy by its own signals.

Path	n	Agentic	Single	Pipeline	Signal	Agentic
direct	186	82	81	81	verified (225)	79
relook	57	39	39	35	partially verified (66)	32
reanswer	35	69	54	57	unsupported/low (9)	11
abstained	22	–	59	50	consistency high (238)	75
					consistency med/low (40)	40–57

answering alone—the check's blind reading—scores 72.0% strict and beats the loop ($p = 0.024$). The loop's mechanisms work individually: revision fixed 23 wrong first answers and broke 5 (92 revised episodes), and when the blind check objected to a first answer it was wrong in 64 of 85 cases. Re-answered cases score 69% versus 54% for the single prompt, but zoomed re-looks, used on the 66 hardest cases, do not beat it (33% vs. 41%). Abstention offsets the gain: in 11 of 22 abstentions the loop's own answer was right, and on the questions it answered it scores 72% versus 69% for the single prompt (Table 2).

Failure case analysis. *Self-consistency:* two independent re-answers per case; high consistency means 75% accuracy, medium/low 40–57%, so low consistency marks uncertainty rather than certain error. *Cross-modal agreement:* because the verifier sees the image, verification status is predictive: verified answers are 79% correct, the rest 29%. *Verifier model:* a larger model verifying a smaller one steers 114 of 300 episodes away from the direct path. *Disagreement:* the three systems disagree on 68% of cases; when exactly one system is right, it is the agentic loop 23 times, the single prompt 24 and the workflow 13. *Stability:* on 100 repeated cases, 9% of single-prompt answers flipped between right and wrong, versus 15% for the agentic loop. Five patterns dominate the errors: (i) a shared perception blind spot, where looker and verifier misread the image the same way; (ii) abstention withholding a right answer when the blind reading disagrees or one image is judged insufficient; (iii) the hardest questions resisting re-looks, with an occasional revision breaking a right answer; (iv) questions the image cannot answer, where abstaining or hedging is scored wrong; and (v) workflow rigidity, where fixed rules answer "Brain" or "X-ray" when a specific finding is needed.

6 User Interface

A React frontend exposes three views: **Dashboard** (upload an image, compare the agentic loop with both baselines, inspect the trace, verification checks and failure categories), **Case Explorer** (browse the 300evaluation cases with full agent traces) and **Evaluation** (strict and judged accuracy for all four systems, accuracy by question type and by controller path, disagreement and a per-case table).

7 Reflection

When agentic behaviour is beneficial. On this benchmark, not for accuracy: the loop matches a single prompt and trails its own verifier answering alone. Its mechanisms help where they act—revision fixes far more answers than it breaks, and the blind check's objections are usually right—but abstention withholds about as many correct answers, and its answers are not more stable between runs. Its benefit is *calibrated confidence*: the verified label separates answers that are 79% correct from the rest at 29%, and every decision is auditable, which suits review workflows that can act on that signal. Answering with a low-confidence flag instead of abstaining, and a cheaper verifier, are the natural next steps.

Limitations. About 59× the cost and 10× the median latency of a single prompt, without an accuracy gain on this benchmark. Abstentions are scored as wrong, so the abstention policy lowers accuracy by design. The verifier shares the perception weaknesses of vision models, so agreement is not correctness, and because its guidance may suggest answers, the loop's contribution cannot be fully separated from the verifier's (a critique-only verifier would isolate it). 300 questions give ±5.3-point intervals and noisy per-type estimates; the LLM judge is itself a model, spot-checked

but not human-annotated. The system uses no retrieval, so it cannot draw on knowledge that is not visible in the image.

References

[1] Lin, Z., Zhang, D., Tao, Q., Shi, D., Haffari, G., Wu, Q., He, M., & Ge, Z. (2023). Medical visual question answering: A survey. *Artificial Intelligence in Medicine*, 143, 102611.

[2] Liu, Y., Wang, Z., Xu, D., & Zhou, L. (2023). Q2ATransformer: Improving medical VQA via an answer querying decoder. In *Information Processing in Medical Imaging – IPMI 2023*, Lecture Notes in Computer Science, vol. 13939, pp. 445–456. Springer.

[3] Liu, B., Zhan, L.M., Xu, L., & Wu, X.M. (2023). Medical visual question answering via conditional reasoning and contrastive learning. *IEEE Transactions on Medical Imaging*, 42(5), 1532–1545.

[4] Qiu, C., Huang, K., Xie, Z., Liu, M., Gu, J., & Zong, X. (2025). Explainable medical visual question answering via chain of evidence. *Knowledge-Based Systems*. doi:10.1016/j.knosys.2025.113672.

[5] Syed, A., et al. (2025). MedVCTP: Improving accuracy and explainability in medical visual reasoning. In *NeurIPS 2025 Workshop on Human-AI Coevolution (HAIC)*.

[6] Ben Abacha, A., Hasan, S.A., Datla, V.V., Liu, J., Demner-Fushman, D., & Müller, H. (2019). VQA-Med: Overview of the medical visual question answering task at ImageCLEF 2019. *Working Notes of CLEF 2019*.

[7] Nguyen, B.D., Do, T.T., Nguyen, B.X., Do, T., Tjiputra, E., & Tran, Q.D. (2019). Overcoming data limitation in medical visual question answering. In *MICCAI 2019*, Lecture Notes in Computer Science, pp. 522–530. Springer.

[8] Zhang, X., Wu, C., Zhao, Z., Lin, W., Zhang, Y., Wang, Y., & Xie, W. (2023). PMC-VQA: Visual instruction tuning for medical visual question answering. *arXiv preprint arXiv:2305.10415*.

[9] Bazi, Y., Al Rahhal, M.M., Bashmal, L., & Zuair, M. (2023). Vision–language model for visual question answering in medical imagery. *Bioengineering*, 10(3), 380.

[10] Li, C., Wong, C., Zhang, S., Usuyama, N., Liu, H., Yang, J., Naumann, T., Poon, H., & Gao, J. (2023). LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day. In *Advances in Neural Information Processing Systems (NeurIPS 2023), Datasets and Benchmarks Track*.

[11] Wu, C., Zhang, X., Zhang, Y., Hui, H., Wang, Y., & Xie, W. (2025). Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data. *Nature Communications*, 16, 7866.

[12] Chen, X., Lai, Z., Ruan, K., Chen, S., Liu, J., & Liu, Z. (2025). R-LLaVA: Improving Med-VQA understanding through visual region of interest. In *ICLR 2025 Workshop on Human-AI Coevolution (HAIC)*. `https://arxiv.org/abs/2410.20327`.

[13] Lau, J.J., Gayen, S., Ben Abacha, A., & Demner-Fushman, D. (2018). A dataset of clinically generated visual questions and answers about radiology images. *Scientific Data*, 5:180251.

[14] Deka, P., Jurek-Loughrey, A., & Deepak, P. (2022). Improved methods to aid unsupervised evidence-based fact checking for online health news. *Journal of Data Intelligence*, 3(4), 474–504. `https://huggingface.co/pritamdeka/S-PubMedBert-MS-MARCO`.